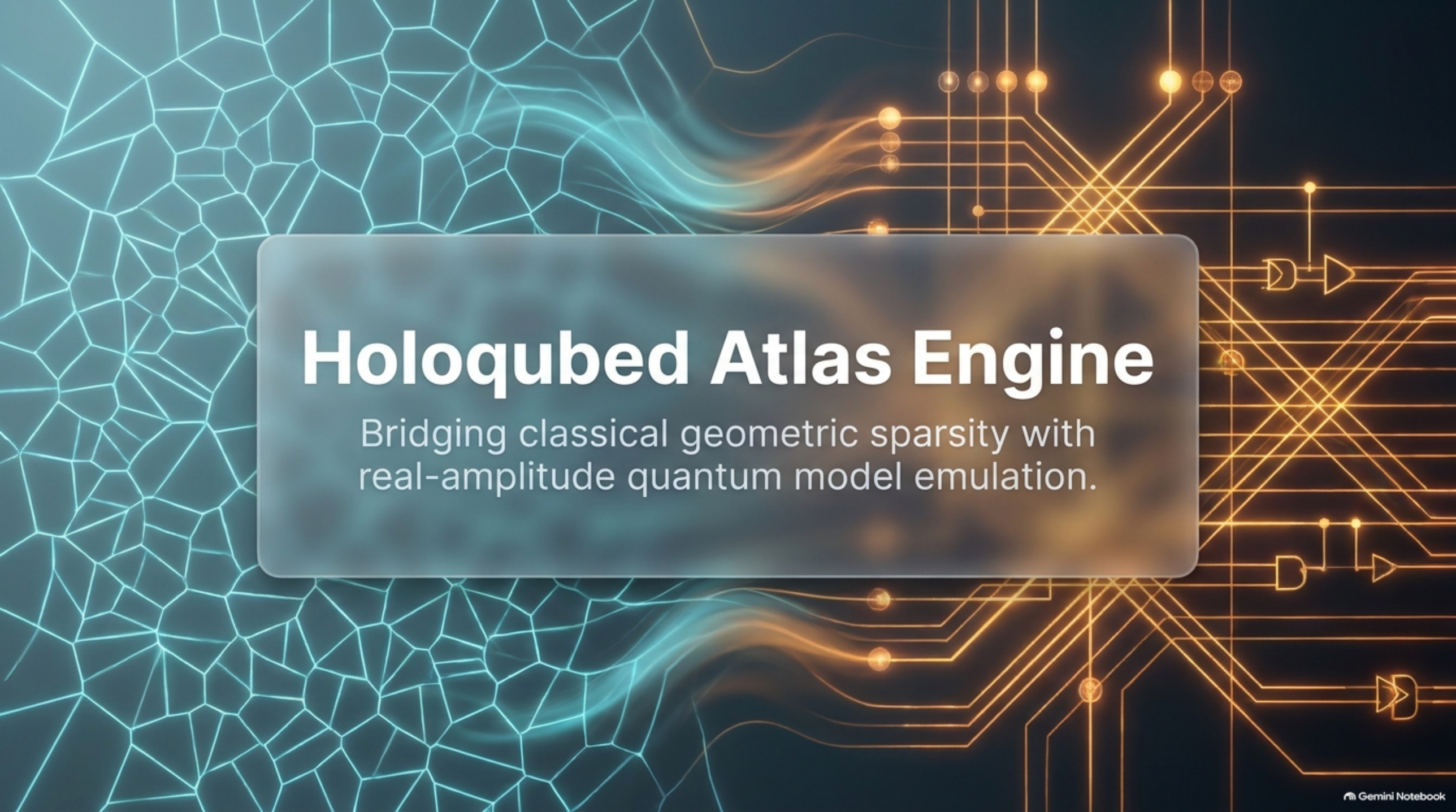
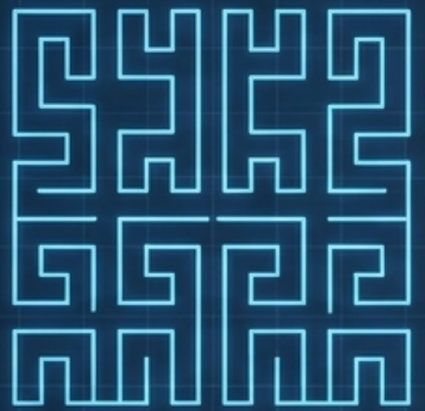
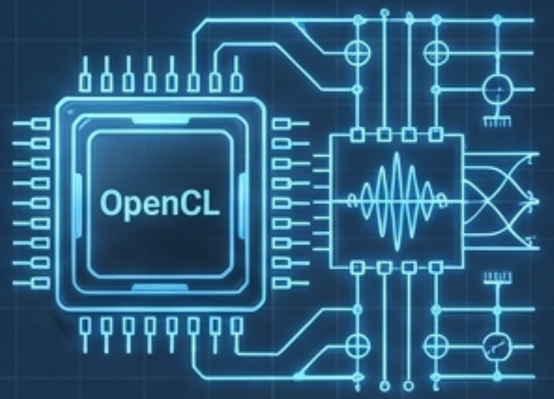

Holoqubed Atlas Engine
Bridging classical geometric sparsity with real-amplitude quantum model emulation.
Gemini Notebook

A comprehensive architecture mapping geometric routing to quantum execution



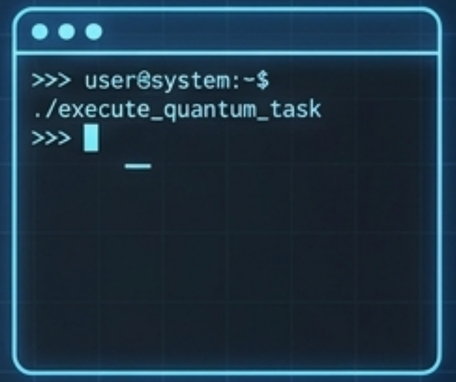
Geometry & Forge
(The Map)

- Voronoi regions
- Exact low-rank coordinates
- Residual ellipsoids



Execution Engine
(The Core)

- OpenCL multi-device round robin
- Dense execution
- PyQrack simulators



Interfaces
(The Output)

- OpenAI-compatible API
- Real-time Live Dashboard
- Landscape analysis

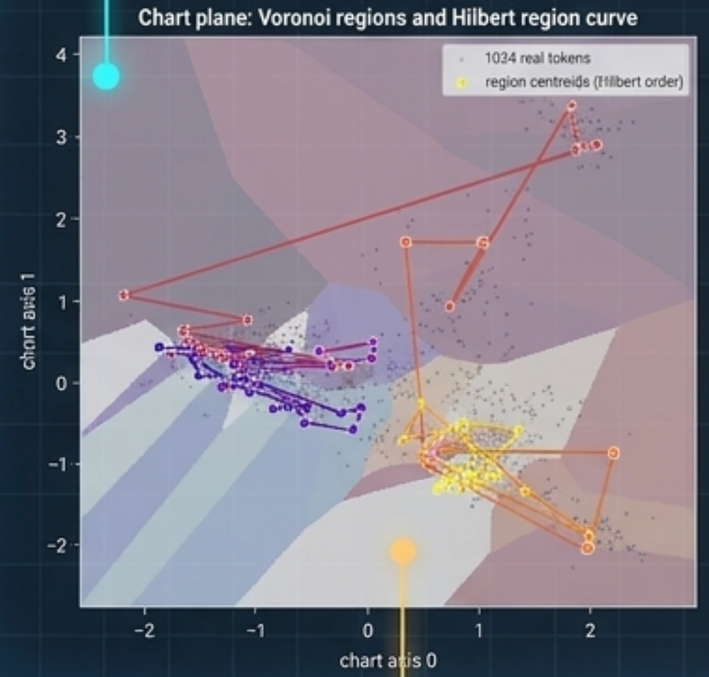
System operates entirely without CUDA or C++ build steps, running locally via OpenCL and Python standard libraries.

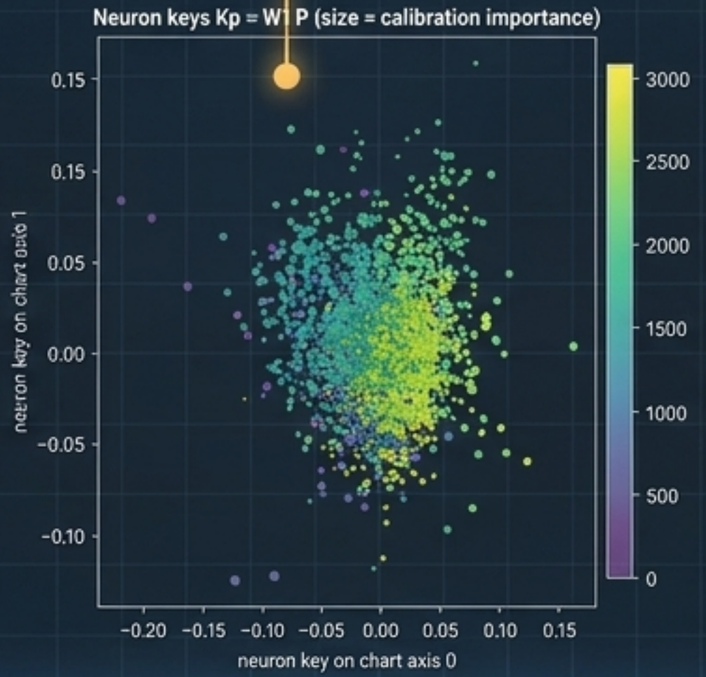
Gemini Notebook

Real-time telemetry proves the viability of geometry-driven local inference

Decode speed: 15.54 tokens/s
(Performant local execution)

MLP work: 32.8%
(Massive computational savings)





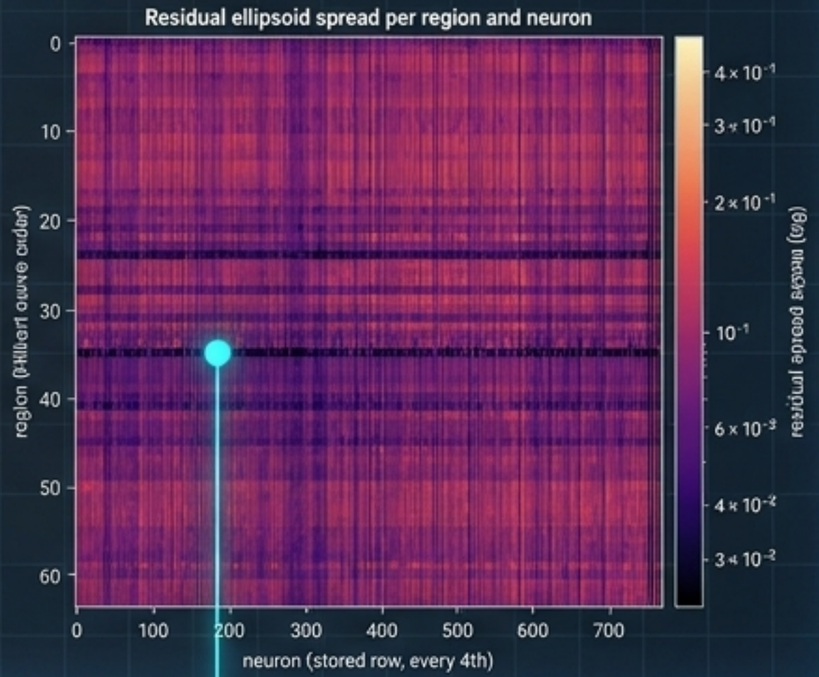


Chart plane
(Live token path tracing
across low-rank space)

Layers over tokens
(Tracing routed neurons per token)

Gemini Notebook

The shape of the MLP input cloud determines which neurons to evaluate

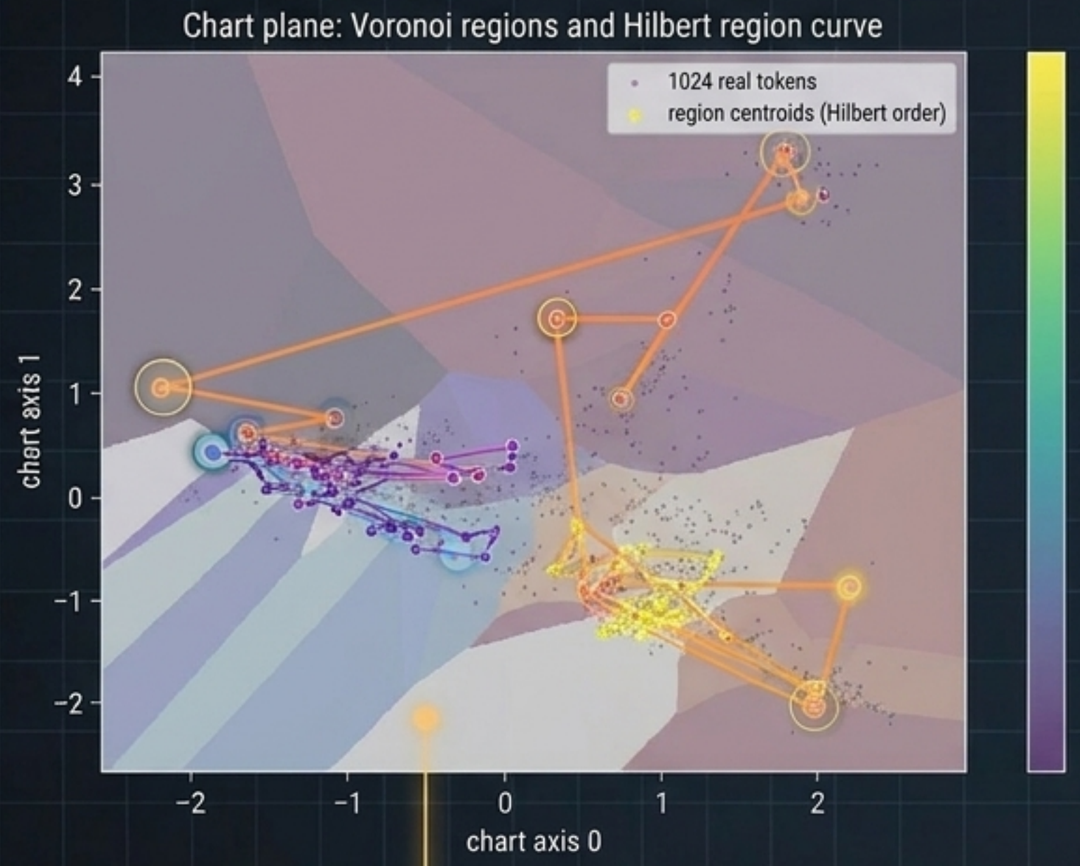


Chart plane
(Live token path tracing
across low-rank space)

THE CONCEPT:

Neurons whose pre-activation cannot plausibly become positive, based on where the token sits on the activation manifold, are mathematically skipped.

THE CHART MATH:

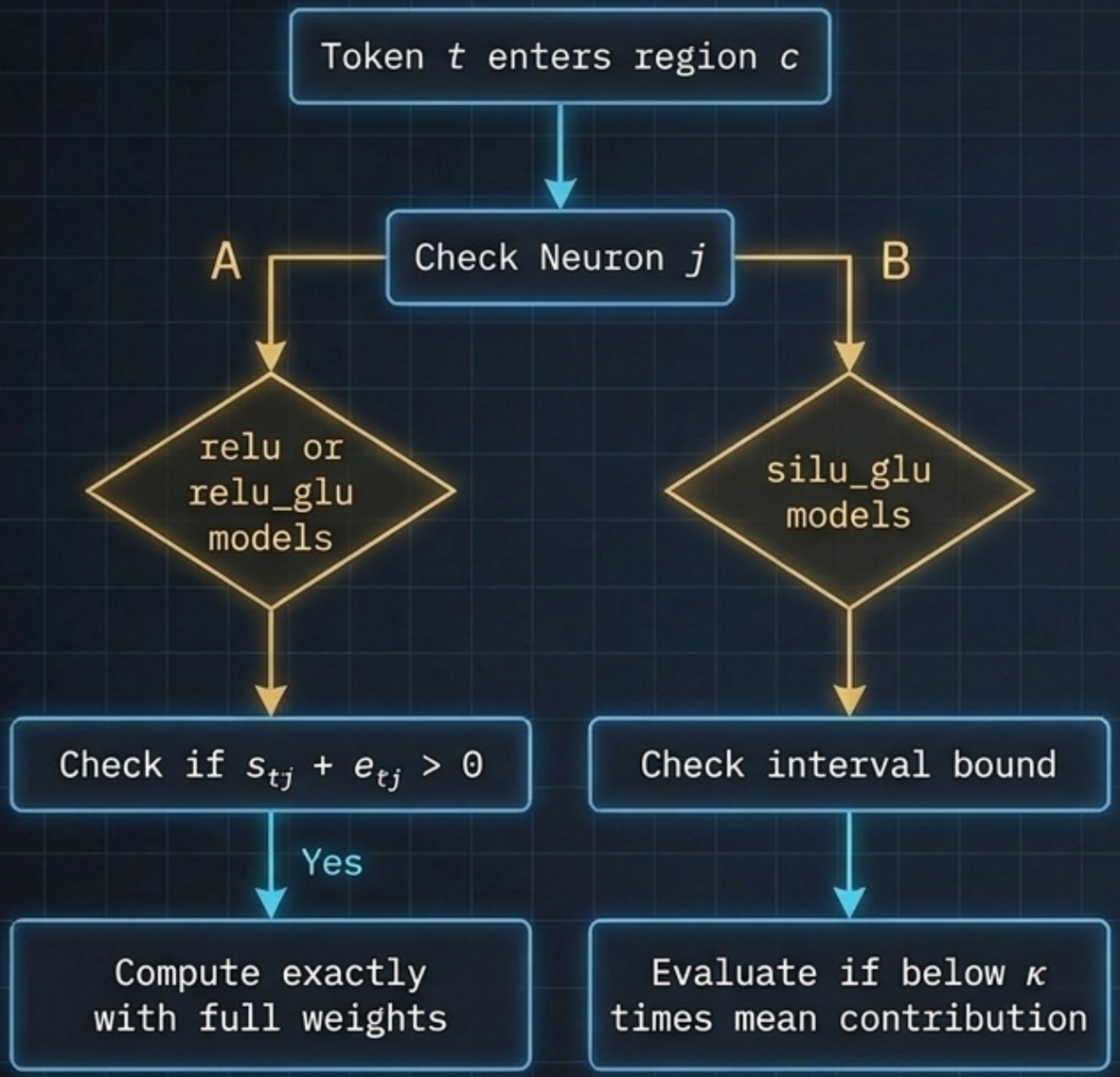
Mean (μ) and principal axes (P) create exact low-rank coordinates: $Z = (x - \mu) P$.

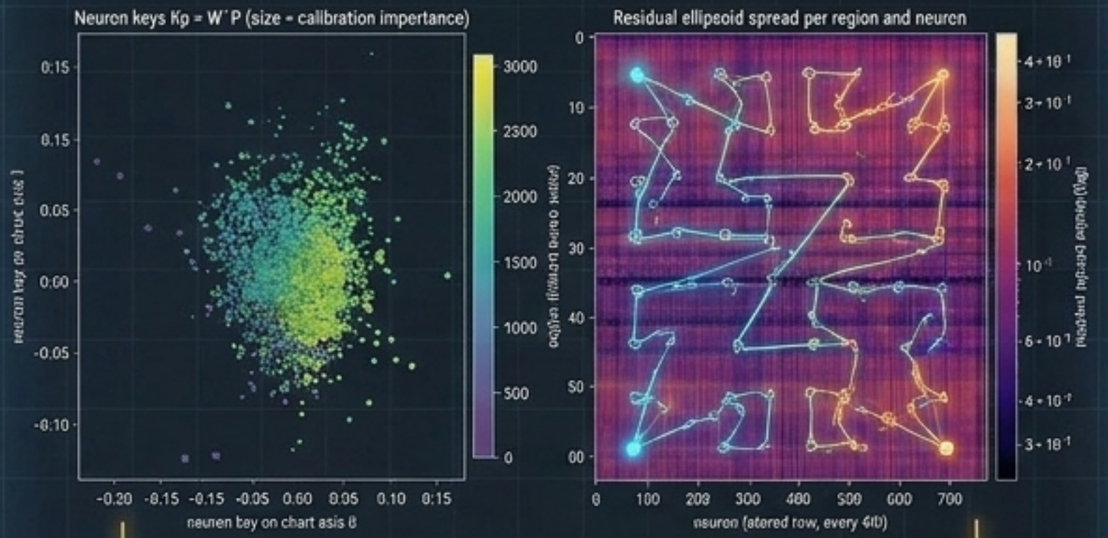
THE RESIDUAL BOUNDS:

Each region possesses a residual ellipsoid based on the RMS of calibration tokens. tokens. If a token stays within its expected radius, bounds hold, and the engine safely ignores inactive neurons.

Gemini Notebook

Hilbert curve reordering maximizes DMA block sharing for evaluated neurons



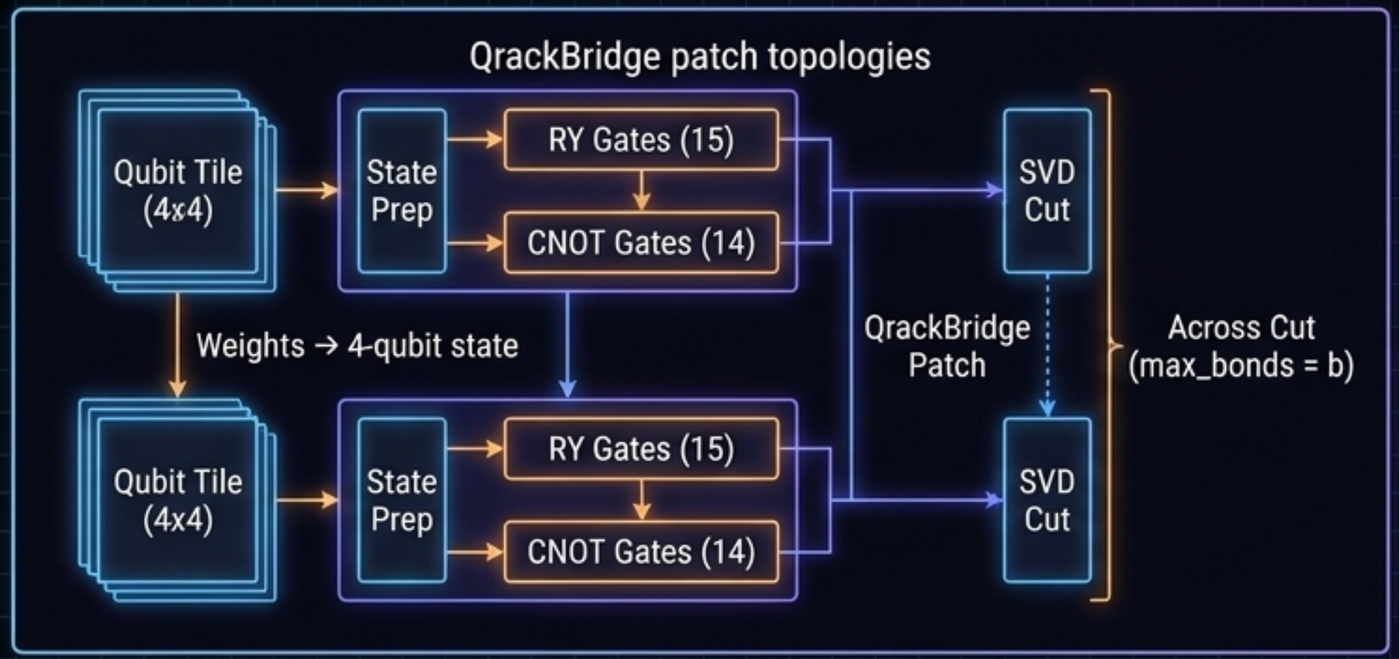


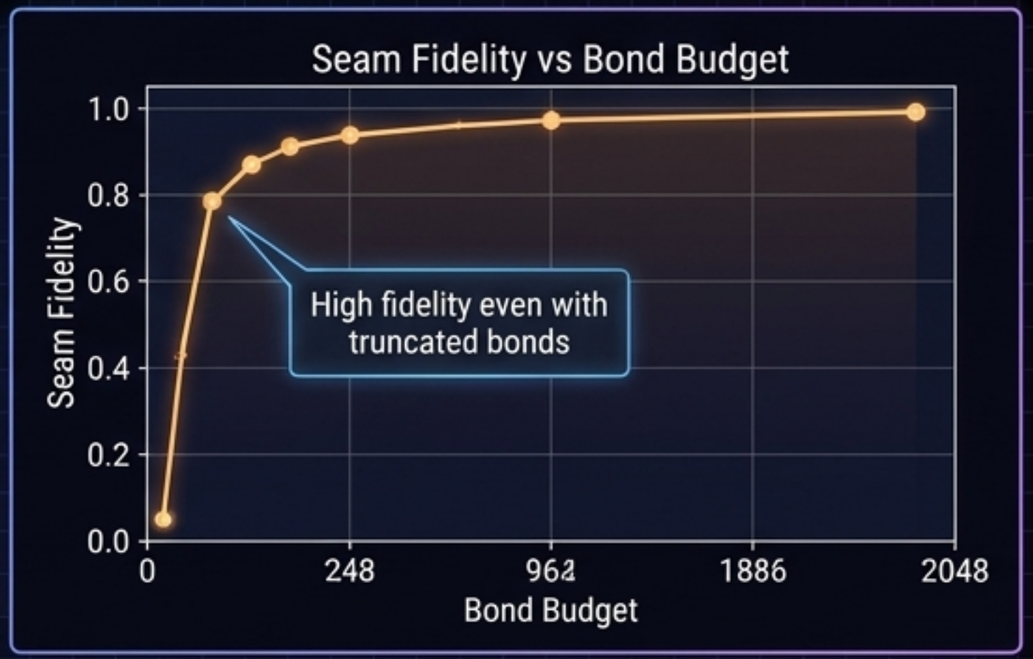
SYT TAKEAWAY:

By reordering neurons along an n-D Hilbert Curve, neurons that fire together share memory blocks, reducing W2T DMA memory reads by 16-17% at z=3.5.

Gemini Notebook

Emulating classical sparsity via real-amplitude state-preparation trees





THE GOAL	THE CONSTRUCTION	THE SEAM CUT
Emulate exact linear maps within quantum statevectors.	Weights are encoded as 4x4 qubit tiles. 16 weights become a 4-qubit state, prepared using 15 RY and 14 single-control CNOT gates.	The QrackBridge limits the SVD rank across the patch cut via a max_bonds = b cap. Fidelity remains exceptionally high even with truncated bond budgets.

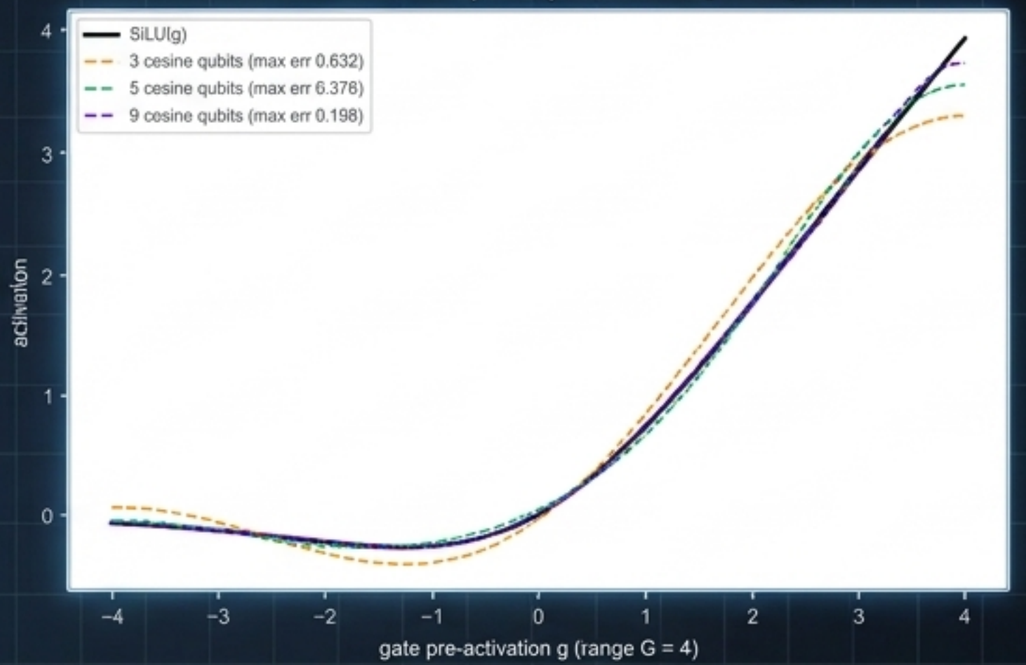
Gemini Notebook

Rebuilding continuous classical activations from discrete qubit expectations

qneuron: 8 gate cosine qubits + 1up qubit (all independent, one layer); h = (c0 + sum c_m <Z_m>) <Z_u> / eps



SiLU rebuilt from qubit expectations cos(m theta)



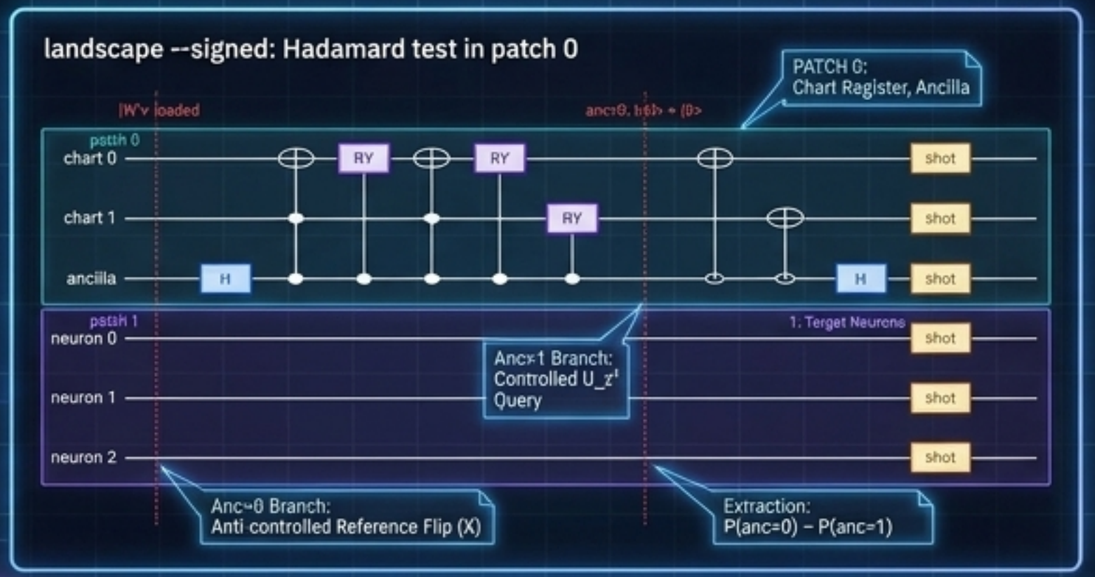
The QrackNeuron translates complex non-linear gating into pure quantum expectations. Gate qubits yield <Z> = cos(mπ(x+1)/2), generating a cosine series fitted to SiLU on a dense grid. and 9 quantumd dass for SiLU on a dense grid. Using just 9 independent cosine qubits plus 1 up qubit reduces the maximum reconstruction error to a negligible 0.198, making whole-model emulation viable.

Gemini Notebook

Hadamard tests extract exact signs from magnitude-only native readouts

1. THE SETUP

A chart register and an ancilla qubit are placed in Patch 0; the target neurons are placed in Patch 1.



2. THE QUERY EXECUTION

A controlled $U_z^\dagger$ query runs on the anc=1 branch. An anti-controlled reference flip runs on the anc=0 branch.

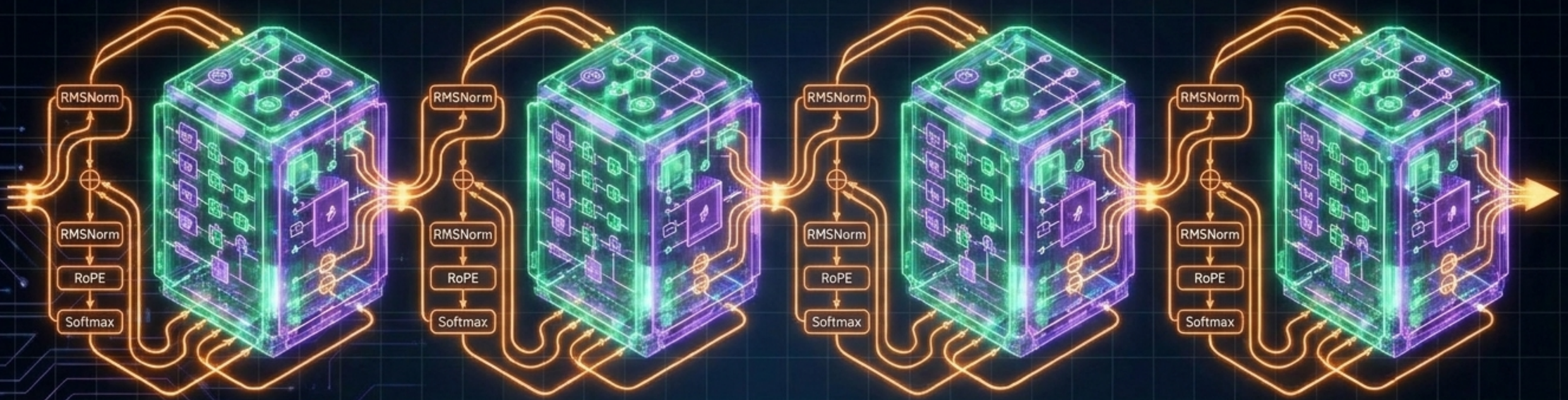
3. THE EXTRACTION

Post-selecting the chart register to $|0\rangle$ yields $P(\text{anc}=0) - P(\text{anc}=1)$.

Simultaneously extracts both sign and magnitude in a single shot batch, achieving near-perfect positive recall (99.2–100%) for active neurons.

Gemini Notebook

Interleaving classical Loom links with fully quantum MLP modules



Total Scale	Qubit Count	The Hybrid Split
37,249,024 4x4 quantum tiles per token.	**148.9 million qubits** if all tiles were resident simultaneously.	Dense nonlinearities (RMSNorm, RoPE, Softmax) execute classically via Loom links. Vast parameter spaces (MLPs) execute via Qrack neurons.

Gemini Notebook

Tree angles at 6 or 8 bits provide the optimal storage-to-precision ratio

Encoding Scheme	Size (MB)	PPL	Top-1 vs Dense	KL Divergence
Baseline BF16	881 MB	28.02	100.00%	0.0000
Gray Angles (Int4)	261 MB	3002 (Failure)	14.90%	5.1142
Tree Angles (Uniform 6/8)	420 MB	28.18	95.88%	0.0054

ANALYSIS: Gray-code leaf angles span 10 radians, causing catastrophic root errors upon quantization. Mottonen Tree Angles divide errors evenly across splits, cutting storage by >50% while preserving a KL divergence of just 0.0054.

Gemini Notebook

Geometric routing yields near-lossless perplexity at a fraction of dense compute

OPT-125m (ReLU) - SUCCESS CASE

36.6%

MLP Work

z = 3.5	46.2% Neurons Routed

PPL: 48.55 (vs 48.37 dense)

Qwen3-0.6B (SiLU GLU) - THE GLU CHALLENGE

108.9%

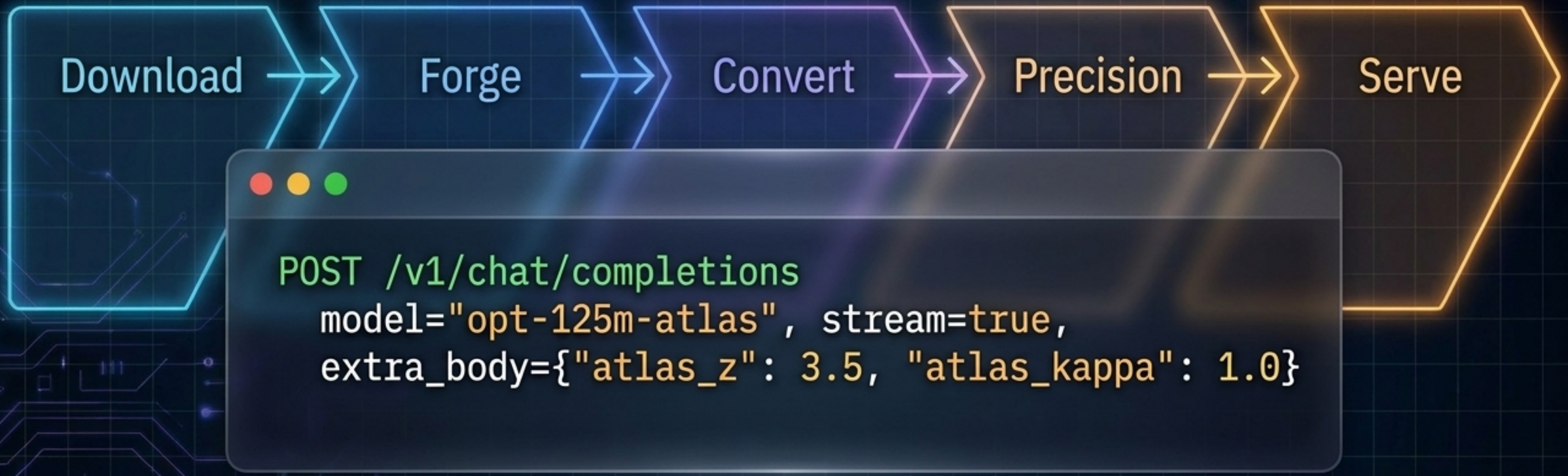
MLP Work

κ = 1.0	Little compute gain realized due to dense activations.

Interval bounds for silu_glu activations only become selective once output quality is degraded. Realizing routing savings on SwiGLU architectures requires explicit ‘ReLU-fication’ of the gate.

Gemini Notebook

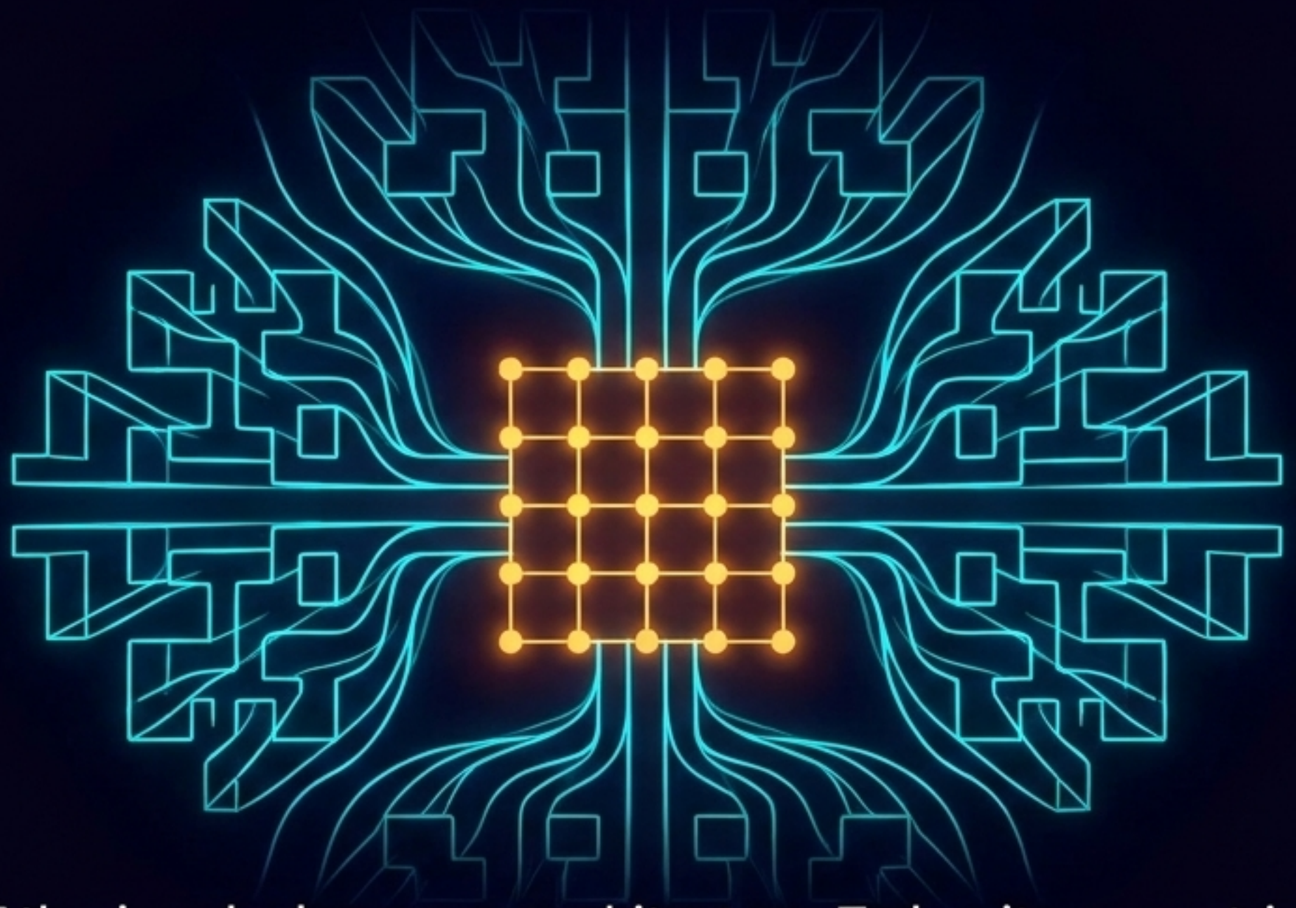
A CUDA-free, OpenAI-compatible pipeline executing entirely on local hardware



The ecosystem is fully integrated via the qwen_holoqubed.sh pipeline. It exposes an OpenAI-compatible SDK endpoint supporting Server-Sent Events (SSE) streaming and per-request geometric routing overrides, operating exclusively on CPU and OpenCL backends.

Gemini Notebook

Bridging immediate classical efficiency with the quantum landscapes of tomorrow



The Holoqubed Atlas is a dual-purpose architecture. Today, its geometric routing delivers immediate memory efficiency and classical sparsity for ReLU networks. Tomorrow, its Qrack integration stands as an exact mathematical testbed, laying the groundwork for executing entire foundation models natively on real-amplitude quantum weight landscapes.

Gemini Notebook